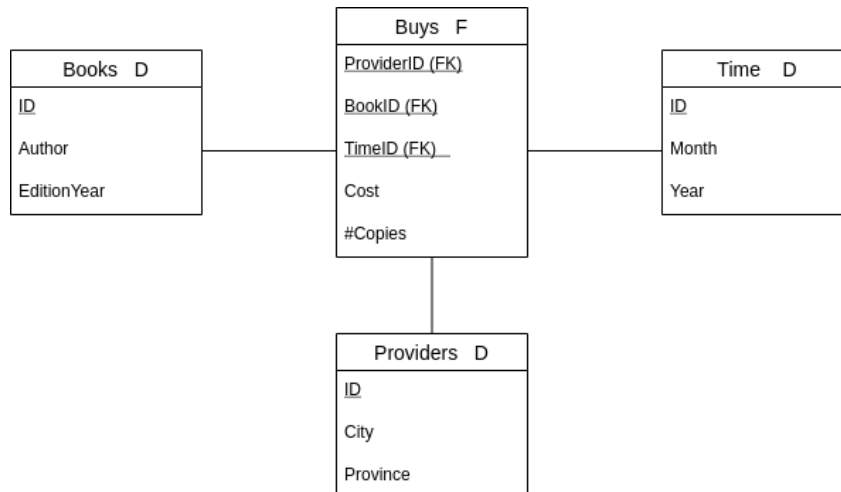


Family name: _____ Given name: _____

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1. Given the multidimensional schema, the sequence of multidimensional operations below, and the **equivalent SQL query**, justify if they correspond to each other and this is the best way to write the query. If they don't correspond or this is not the best way to write it, briefly explain why and how the SQL query should be modified.



- A:=Roll-up(Buys, Time.Month, Sum)
- B:=Selection(A, Month="2023-10")
- C:=Roll-up(B, Providers.AllProviders, Sum)
- D:=ChangeBase(C, {Time, Books})
- E:=Roll-up(D, Time.AllDays, Sum)
- R:=Projection(E, Cost)

```

SELECT "AllDays", bo.ID, SUM(bu.Cost)
FROM Buys bu, Books bo
WHERE bo.ID=bu.BookID
GROUP BY bo.ID
ORDER BY bo.ID;
  
```

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2. Create a **multidimensional logical design** (i.e., relational tables and the corresponding constraints) for the following CSV file, assuming that **CountPlayers** indicates the number of players whose data are represented in the row, and **SumX** their statistics. Explicit any assumption you make.

Team	Gender	AgeRange	PlayedEuroleague	Month	SumPoints	SumRebounds	SumFaults	CountPlayers	SumMinutes
Barcelona	M	20-30	T	Oct23	146	57	40	4	300
Barcelona	M	20-30	F	Oct23	30	3	2	1	30
Barcelona	M	-20	F	Oct23	7	1	1	1	5
Lyon	F	-20	F	Oct23	5	7	3	2	17
Avenida	F	+30	T	Sep22	10	2	4	1	20
Joventut	M	-20	F	Dec22	8	2	3	1	9

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3. Transform the following **run-length encoding** into a matrix of bits where rows correspond to tuples and columns to values. Briefly **justify** whether it is correct or not.

**Run-length
encoded**

x	y	z	t
1	1	0	1
0	0	0	1
1	1	1	0
1	0	0	1
1		1	0
0		1	1
1			0
1			0

⇒

Bits matrix

x	y	z	t

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4. Given the following schema, create the **syntactic tree** corresponding to the query¹ and optimize it. Only the final optimized tree needs to be provided.

Schema:

Subjects(IDSub, nameSub, credits)

Students(IDStu, nameStu, degree, entryYear)

Registration2005(IDStu, IDSub, mark, repeat)

- {IDStu} FK to Students
- {IDSub} FK to Subjects

Registration2004(IDStu, IDSub, mark, repeat)

- {IDStu} FK to Students
- {IDSub} FK to Subjects

Query:

```
SELECT s.IDStu, s.nameStu, r.mark
FROM   Students s,
      (
        SELECT * FROM Registration2004
        EXCEPT
        SELECT * FROM Registration2005) r
WHERE  r.repeat='Y' AND s.entryYear=2000 AND s.IDStu=r.IDStu;
```

¹EXCEPT is the SQL clause corresponding to set difference.